

Unit 5.3 The definite integral

Note Title

2012/05/11

Section 5.3

Definition of a Definite Integral p 872

If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = \frac{b-a}{n}$. We let $x_0 (= a), x_1, x_2, \dots, x_n (= b)$

be the end points of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any sample points in these subintervals, so x_i^* lies in the i th subinterval $[x_{i-1}, x_i]$.

Then the definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is integrable on $[a, b]$.

$f(x)$ is called the integrand.

a and b are called the limits of integration.

When do we know that $\int_a^b f(x) dx$ exists?

Theorem on p 373

If f is continuous on $[a, b]$ or f has a finite number of jump discontinuities on $[a, b]$ then $\int_a^b f(x) dx$ exists.

The relationship between area between a curve and the x -axis and definite integral.

- If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x) \geq 0$ because it is equal to the area of the region between the graph and the x -axis and $x=a$ and $x=b$.
- If $f(x) \leq 0$ on $[a, b]$ then $\int_a^b f(x) \leq 0$ because it is equal to the negative

of the area of the region between the graph and the x -axis and $x=a$ and $x=b$.

- If $f(x) \geq g(x)$ on $[a, b]$ then
$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

Properties of the definite integral.

$$1. \int_b^a f(x) dx = - \int_a^b f(x) dx$$

$$2. \int_a^a f(x) dx = 0$$

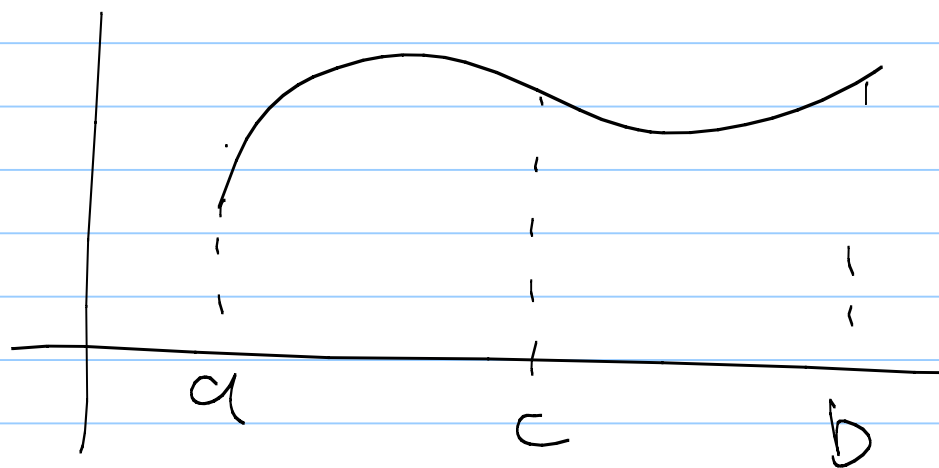
$$3. \int_a^b c dx = c [b - a]$$

$$4. \int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$5. \int_a^b c f(x) dx = c \int_a^b f(x) dx$$

$$6. \int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

$$7. \int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$



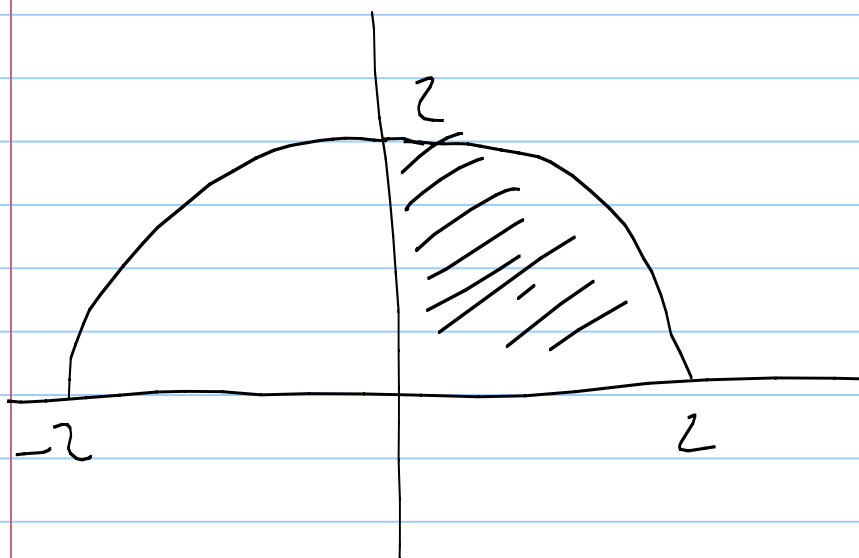
Examples

Use the area to calculate the following integrals

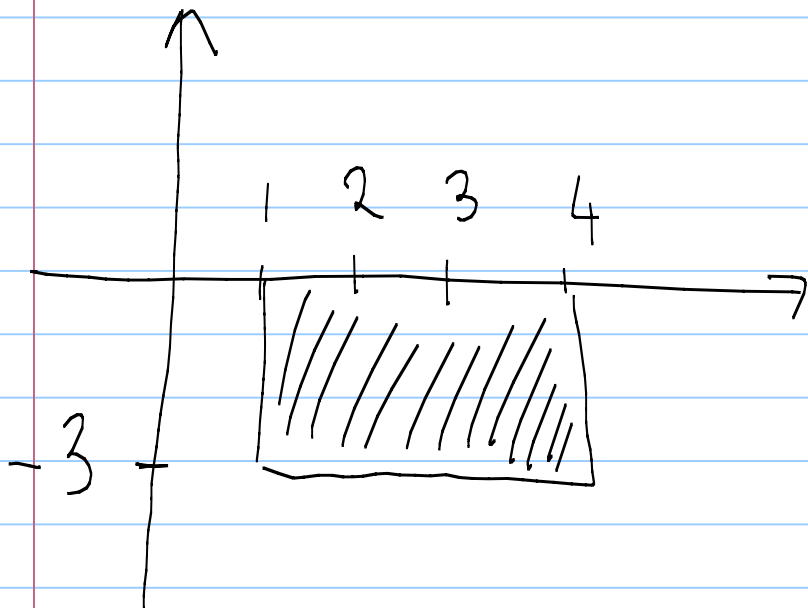
1. $\int_0^2 \sqrt{4 - x^2} \, dx$

$$\begin{aligned} \text{doc} &= \frac{1}{4} \pi r^2 \\ &= \frac{1}{4} \pi (2)^2 \end{aligned}$$

$$= \pi$$



$$2. \int_1^4 -3 \, dx = 3x - 3 \\ = -9$$



3)

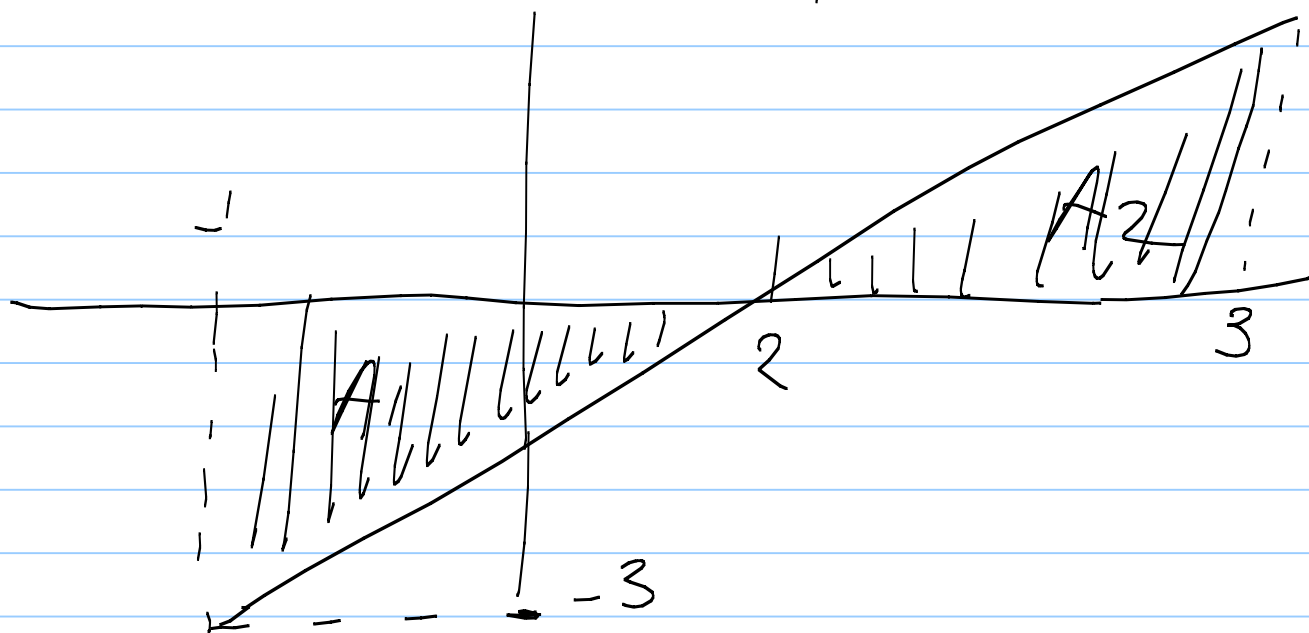
$$\int_{-1}^3$$

$$x-2 \, dx$$

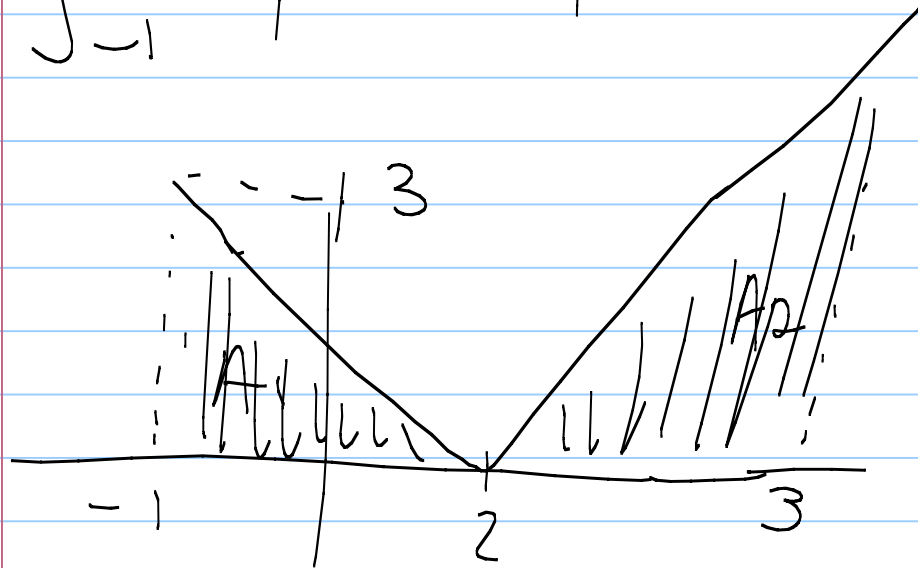
$$= -A_1 + A_2$$

$$= -\frac{1}{2} [3 \times 3] + \frac{1}{2} [1 \times 1]$$

$$= -4$$

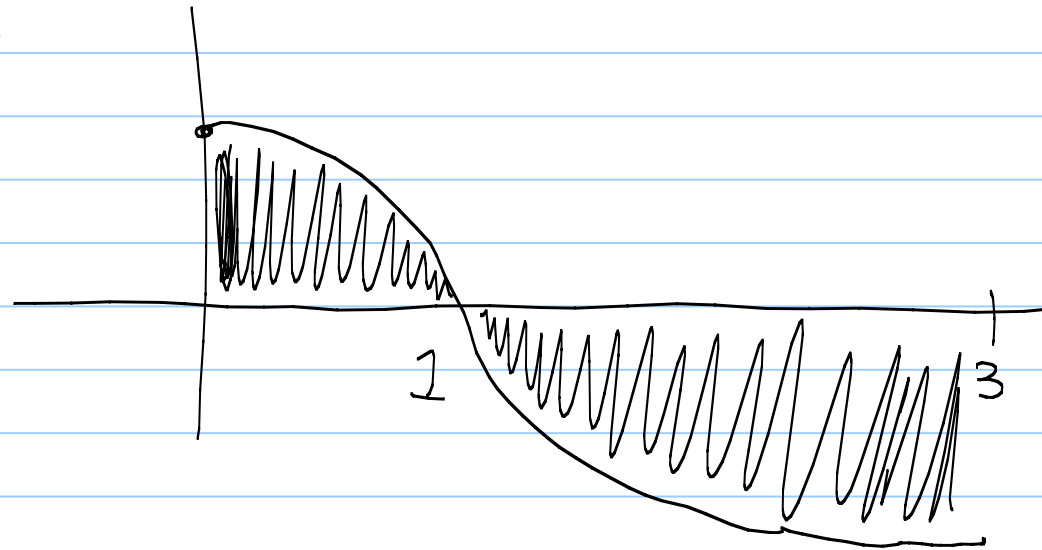


4) $\int_{-1}^3 |x-2| dx$



$$\begin{aligned} \int_{-1}^3 |x-2| dx &= A_1 + A_2 \\ &= \frac{1}{2} [3 \times 3] + \frac{1}{2} [1 \times 1] = 5 \end{aligned}$$

Example Consider the shaded area of f



Express the area in terms of definite integrals.

Sol :

$$\int_0^1 f(x) dx - \int_1^3 f(x) dx$$

Example. If $\int_0^{10} f(x) dx = 17$

and $\int_0^8 f(x) dx = 12$, Find $\int_8^{10} f(x) dx$

Sol: $\int_0^8 f(x) dx + \int_8^{10} f(x) dx = \int_0^{10} f(x) dx$

So $\int_8^{10} f(x) dx = \int_0^{10} f(x) dx - \int_0^8 f(x) dx$

$$= 17 - 12$$

$$= 5.$$